Blessed G/Michael Catholic Secondary School

Semester 1 Agriculture Mid1 Examination for Grade11

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

Name: _____ **Grade & Sec:** Grade11 _____ **R.No:** _____

I. Choose the best answer among the given alternatives.

____ **1.** Which soil horizon contains the highest concentration of organic matter and humus?

A. A horizon (Topsoil) **B.** B horizon (Subsoil)

C. C horizon (Parent material) **D.** R horizon (Bedrock)

____ **2.** Which of the following is an effective soil conservation practice on sloped agricultural terrain?

A. Contour plowing and terracing **B.** Downslope cultivation

C. Overgrazing **D.** Deforestation

____ **3.** Which process in leguminous crops enriches soil with bioavailable nitrogen compounds?

A. Nitrification by Rhizobium root nodules
B. Denitrification by Pseudomonas
C. Ammonification by fungi
D. Photosynthesis by stomata

____ **4.** Soil texture is defined by the relative proportions of which three particle fractions?

A. Sand, silt, and clay **B.** Gravel, humus, and water

C. Nitrogen, phosphorus, and potassium **D.** Loam, limestone, and basalt

____ **5.** Which soil property measures the relative acidity or alkalinity of the soil solution?

A. Soil pH **B.** Cation exchange capacity (CEC)

C. Soil bulk density **D.** Soil porosity

____ **6.** Which traditional organic amendment improves soil structure, aeration, and moisture retention while returning organic matter?

A. Compost and farmyard manure **B.** Synthetic urea granules

C. Copper sulfate spray **D.** Hydrated lime powder

____ **7.** Which agricultural practice alternates different crop families on the same parcel of land across successive planting seasons?

A. Crop rotation **B.** Monoculture

C. Slash and burn **D.** Continuous cropping

____ **8.** In Ethiopia, the Vertisol (black clay soil) is characterized by which physical management challenge?

A. Severe swelling when wet and deep cracking when dry, causing poor drainage
B. Extremely sandy texture that cannot hold water
C. Complete absence of iron minerals
D. Permanent frozen permafrost

____ **9.** What is the primary role of nitrogen (N) in crop nutrition?

A. Promotes vigorous vegetative leafy growth and chlorophyll synthesis
B. Stimulates root nodulation exclusively
C. Accelerates fruit ripening only
D. Prevents soil salinity

____ **10.** Which soil amendment is commonly applied to neutralize strongly acidic soils and increase pH?

A. Agricultural limestone (Calcium carbonate, $CaCO3$)
B. Sulfur powder
C. Ammonium sulfate fertilizer
D. Phosphoric acid

____ **11.** The agricultural practice of planting trees or woody shrubs alongside agricultural crops or livestock is known as:

A. Agroforestry **B.** Monoculture

C. Silviculture exclusively **D.** Deforestation

____ **12.** Which of the following is an indigenous Ethiopian cereal grain prized for high iron content and gluten-free flour?

A. Teff (Eragrostis tef) **B.** Barley (Hordeum vulgare)

C. Wheat (Triticum aestivum) **D.** Rye (Secale cereale)

____ **13.** What is the main advantage of minimum (conservation) tillage over conventional deep plowing?

A. Reduces soil erosion and preserves soil structure and soil moisture

B. Increases fuel consumption

C. Completely eradicates all weed seeds instantly

D. Increases soil compaction

____ **14.** Which soil organism is celebrated as a "natural soil aerator" by burrowing channels and cycling organic residues into fertile casts?

A. Earthworms **B.** Termites exclusively

C. Nematodes **D.** Aphids

____ **15.** What is the term for the artificial application of controlled amounts of water to agricultural land to assist crop production during dry periods?

A. Irrigation **B.** Infiltration **C.** Transpiration **D.** Percolation

____ **16.** Which deficiency symptom is typical of phosphorus deficiency in young crop leaves?

A. Purplish or reddish discoloration along leaves and stunted root growth

B. Interveinal chlorosis only

C. Leaf curling with necrotic spots

D. Excessive lush vegetative growth

____ **17.** The layer of living plant material that protects the soil surface and is incorporated into the soil while still green to enrich fertility is called:

A. Green manure **B.** Inorganic fertilizer

C. Topsoil cover **D.** Slag

____ **18.** What is mulching in horticultural and crop production?

A. Covering the soil surface with organic or synthetic material to conserve moisture and suppress weeds

B. Applying pesticide via aerial spray

C. Flooding fields with saline water

D. Burning crop stubble

____ **19.** Biological pest control involves using:

A. Natural predators, parasites, or pathogens to manage pest populations

B. Broad-spectrum chemical organophosphates

C. Flamethrowers on crop fields

D. Electric fences exclusively

____ **20.** The water held in the soil pore spaces between field capacity and permanent wilting point that plants can absorb is termed:

A. Available soil water **B.** Gravitational water

C. Hygroscopic water **D.** Capillary runoff